

LAB # 12

Searching for Elements in an Array using MIPS Assembly Language

Objective:

To implement a searching operation in MIPS assembly language that allows the program to locate a specific element in an array using loops and comparison instructions.

Theory:

Searching is a basic operation performed on arrays. In MIPS, arrays are handled using memory addresses, loop counters, and comparison instructions.

1. Searching:

Searching is the process of checking the elements stored in a list or array to find whether a particular value exists or not. It compares the target value with each stored element until it is found or until the end of the list is reached. Searching helps us quickly locate data inside memory.

For Searching We Implement in MIPS:

- Loading the base address of the array
- Using loop counters to move through the array
- Comparing values using branch instructions
- Increasing the address by 4 bytes to reach next element

SAMPLE PROGRAM:

```
.data
```

```
arr: .word 5, 9, 2, 8, 3
```

```
msg: .asciiz "Enter a Number: "
```

```
msg2: .asciiz " At index no: "
```

```
msg_f: .asciiz "Found: "
```

```
msg_nf: .asciiz "Not Found"
```

```
.text
```

```
.globl main
```

```
main:
```

```
    la $t0, arr      # array start
```

```
    li $t1, 5        # size
```

```
    li $t4, 0        # index counter = 0
```

```
    li $v0, 4
```

```
    la $a0, msg
```

```
    syscall
```

```
    li $v0, 5        # read number
```

```
    syscall
```

```
    move $t2, $v0     # t2 = key
```

```
loop:
```

```
    beqz $t1, not_found
```

```
    lw $t3, 0($t0)    # load array element
```

```
    beq $t3, $t2, found # if match → found
```

```
    addi $t0, $t0, 4  # next element
```

```
addi $t1, $t1, -1    # size--  
addi $t4, $t4, 1     # index++  
j loop
```

found:

```
li $v0, 4  
la $a0, msg_f  
syscall
```

```
move $a0,$t3         # print found value  
li $v0,1  
syscall
```

```
li $v0, 4  
la $a0, msg2         # "At index no: "  
syscall
```

```
move $a0,$t4         # print index number  
li $v0,1  
syscall  
j exit
```

not_found:

```
li $v0, 4  
la $a0, msg_nf  
syscall
```

exit:

```
li $v0, 10
```

```
syscall
```

OUTPUT:

```
Enter a Number: 3
Found: 3 At index no: 4
-- program is finished running --
```

Lab Tasks:

1. Write a MIPS program in which the user enters 5 integers, stores them in an array, then enters 1 number to search. the program should print whether the number is found or not found, and if found, show the index number.
2. Write a MIPS program to search and find the maximum value inside an array of 7 integers entered by the user. After processing, display the maximum value and the index where it was found.
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